Optimising Digital Advertising Placement Using Predictive Modelling and Contextual Decision Policies

Raneem Belbisi

Abstract

This investigation examines how predictive modelling and contextual decision policies can be used to optimise digital advertising placement by maximising expected conversions per impression (ECPI). While industry practice typically evaluates click-through rate and conversion rate independently, this study reframes the optimization problem through ECPI as a unified metric that captures the full effectiveness of an advertisement impression.

A complete analytical pipeline was developed using a structured advertising dataset, incorporating preprocessing, feature engineering, exploratory analysis, predictive modelling, and policy evaluation. A gradient boosting model was trained to estimate ECPI based on demographic, contextual, and advertisement-related variables. The model achieved a very high level of predictive accuracy, indicating that the relationships within the dataset are highly structured.

To evaluate decision-making performance, a contextual allocation policy was implemented and compared against baseline strategies, including random, most frequent, and global best placement selection. The model-based policy consistently achieved the strongest performance among feasible strategies. However, the magnitude of improvement over baseline approaches remained modest, suggesting that variation in ECPI across placement options is inherently limited. At the same time, the gap between the model-based policy and the oracle benchmark indicates that additional predictive signals remain unexploited.

An interactive dashboard was developed to operationalise the analytical pipeline, enabling users to explore performance across different contextual conditions. This demonstrates the importance of combining predictive modelling with interpretable visualisation tools to support practical decision-making.

Overall, the findings show that predictive modelling can improve advertising placement decisions, but only to a bounded extent. The results suggest that optimisation performance is constrained less by model choice and more by the underlying signal strength and variability of the data, implying that improvements in data quality may yield greater benefits than further increases in model complexity.

Table of Contents

1. Introduction…….. 4
2. Background……..5
3. Work Undertaken……..7
4. Outputs…….. 15
5. Further Work…….. 28
6. Conclusion…….. 29
7. References……..30
8. Appendices……..30

1. Introduction

Digital advertising has become a central mechanism through which organisations reach and influence consumers in online environments. Platforms such as social media networks, search engines, and websites deliver billions of advertisement impressions daily, creating a highly competitive ecosystem in which advertisers must continuously optimize their strategies to remain effective. Within this environment, determining where an advertisement should be placed is not a trivial decision. Advertising performance varies across contexts, influenced by factors such as user demographics, advertisement characteristics, timing, and platform. As a result, selecting the most effective placement requires more than simple intuition or static rules, and increasingly depends on data-driven decision-making.

Advertising effectiveness is commonly evaluated using metrics such as click-through rate (CTR) and conversion rate. CTR reflects the proportion of impressions that result in user engagement, while conversion rate captures the likelihood that a click leads to a desired outcome. Although these measures provide useful insights individually, they do not directly represent the overall value of an advertisement impression. In practice, advertisers are concerned with the probability that a single impression leads to a successful conversion. This is captured by Expected Conversions per Impression (ECPI), defined as the product of CTR and conversion rate. By combining engagement and outcome into a single metric, ECPI provides a more comprehensive measure of advertising effectiveness and is particularly suitable for optimisation tasks.

Recent advances in data science have enabled the use of predictive modelling to analyse large-scale advertising data and identify relationships between contextual variables and performance outcomes. Unlike traditional approaches that rely on aggregate trends or fixed heuristics, predictive models allow performance to be estimated at a more granular level, taking into account multiple interacting factors simultaneously. This creates the opportunity to move from descriptive analysis towards decision-oriented systems, where predictions are used to guide the selection of advertisement placements that maximise expected outcomes.

This project investigates the extent to which predictive modelling, when combined with contextual decision policies, can improve digital advertising placement decisions. Using a structured dataset containing demographic, contextual, and advertisement attributes, an end-to-end analytical pipeline was developed. This pipeline includes data preprocessing, feature engineering, exploratory analysis, predictive modelling, and policy evaluation. A new target variable, ECPI, is constructed to represent the expected effectiveness of an advertisement impression, forming the basis for both modelling and decision-making.

Feature engineering plays a key role in capturing patterns that influence advertising performance. Temporal features derived from timestamp data, such as hour of day and day of week, are used to reflect behavioural variation over time. Demographic variables are grouped into meaningful categories, including age bands and income levels, to reduce noise and improve interpretability. These transformations allow the model to capture relationships that are not immediately visible in raw data and provide a more structured representation of the advertising environment.

A gradient boosting regression model is then trained to predict ECPI based on the engineered features. This modelling approach is well suited to structured tabular data and is capable of capturing non-linear relationships between variables. By estimating the expected performance of advertisement impressions across different contexts, the model provides a foundation for informed placement decisions.

However, predictive accuracy alone does not guarantee improved decision-making. To address this, the project incorporates a policy evaluation framework that assesses how predictions translate into actionable outcomes. A model-based policy selects the advertisement placement predicted to maximise ECPI for each context, and its performance is compared with baseline strategies including random selection, global best selection, and most frequent placement. An oracle benchmark is also included to represent the theoretical upper bound of achievable performance, providing a reference point for evaluating the effectiveness of the model.

The central research question guiding this investigation is:

To what extent can predictive modelling combined with contextual decision policies improve digital advertising placement decisions by maximising expected conversions per impression?

To answer this question, the project develops and evaluates a complete analytical framework that links data processing, predictive modelling, and decision-making. The results are presented through both analytical outputs and an interactive dashboard, which allows users to explore performance patterns dynamically across different contexts. By integrating predictive modelling with decision policies, this study demonstrates how machine learning can be applied not only to understand advertising performance, but also to improve practical decision-making in digital advertising environments.

2. Background

Digital advertising has become one of the dominant forms of marketing in the modern digital economy. Organisations increasingly allocate large portions of their marketing budgets to online platforms such as search engines, websites, and social media networks, where advertisements can be delivered to highly targeted audiences. The rapid growth of these platforms has generated vast amounts of behavioural and contextual data describing how users interact with advertisements. This data enables organisations to evaluate advertising performance and optimise campaign strategies using data-driven approaches (Chaffey & Ellis-Chadwick, 2019). As a result, digital advertising analytics has emerged as an interdisciplinary field combining marketing, data science, and machine learning techniques to improve advertising effectiveness.

One of the central challenges in digital advertising is measuring and optimising advertisement performance. Two widely used metrics in digital marketing analytics are click-through rate (CTR) and conversion rate. CTR measures the proportion of advertisement impressions that lead to user clicks, providing an indication of how attractive or engaging an advertisement is to viewers. Conversion rate measures the proportion of advertisement clicks that result in a desired

outcome such as a purchase, subscription, or registration (Richardson et al., 2007). While both metrics provide useful insights, analysing them independently does not fully represent the overall value generated by an advertisement impression.

To address this limitation, performance metrics that combine engagement and outcome probabilities are often used in advertising analytics. One such metric is expected conversions per impression (ECPI), which can be defined as the product of click-through rate and conversion rate. ECPI represents the expected probability that a single advertisement impression ultimately results in a successful conversion. By combining engagement probability with post-click success probability, ECPI provides a more comprehensive measure of advertisement effectiveness and is particularly useful for optimisation tasks in digital marketing environments.

The availability of large-scale advertising datasets has also encouraged the use of machine learning techniques for predicting advertising outcomes. Predictive models have been widely applied to tasks such as click prediction, conversion prediction, and advertisement performance forecasting. Early research demonstrated that user behaviour and contextual variables can be used to estimate click probabilities and improve advertisement targeting (Richardson et al., 2007). More recent studies have shown that machine learning algorithms such as gradient boosting, logistic regression, and deep learning models are capable of capturing complex relationships between user demographics, advertisement characteristics, and contextual features that influence advertising outcomes (Zhang et al., 2014).

Among these techniques, gradient boosting models have become particularly popular for structured advertising datasets because they can model nonlinear relationships while maintaining strong predictive performance. Gradient boosting methods build ensembles of decision trees that iteratively improve prediction accuracy by correcting errors from previous models (Friedman, 2001). These methods have been successfully applied across a wide range of predictive analytics problems, including marketing analytics and advertising performance modelling.

While predictive modelling can estimate expected advertisement performance, prediction alone does not directly solve the decision problem faced by advertisers. In practice, advertisers must determine which advertisement placement or channel should be selected in order to maximise campaign outcomes. This introduces an optimisation problem that requires combining predictive estimates with decision policies capable of selecting the most effective action in a given context.

Research in recommender systems and contextual decision models provides useful approaches for addressing such optimisation problems. Recommendation systems aim to identify the most appropriate option for a given user or context by analysing historical interaction data and predicting expected outcomes. Context-aware recommender systems extend this concept by incorporating contextual information such as user demographics, time, and location into the recommendation process (Adomavicius & Tuzhilin, 2015). In digital advertising, similar approaches can be used to recommend advertisement placements that maximise expected engagement or conversion outcomes.

However, an important challenge in applying predictive models to decision systems is evaluating whether model-based decisions actually improve outcomes compared with simpler strategies. Predictive accuracy does not necessarily guarantee improved decision performance. As a result, comparing multiple decision policies has become an important component of applied machine learning systems. Policy evaluation frameworks allow researchers to assess the effectiveness of different decision strategies, such as model-based recommendations, random selection, or heuristic approaches based on historical frequencies (Sutton & Barto, 2018).

Although existing research has extensively explored click prediction, conversion modelling, and recommendation systems, most studies treat prediction and decision-making as separate problems. Many models focus on improving predictive accuracy without evaluating how these predictions translate into practical decision outcomes. In addition, much of the literature evaluates performance at an aggregate level rather than within a contextual decision framework where individual actions must be selected under uncertainty. As a result, there remains a gap in understanding how predictive models can be integrated with decision policies to directly optimize advertisement placement in real-world settings. This project addresses this gap by combining predictive modelling with policy evaluation, allowing both predictive performance and decision effectiveness to be assessed within a unified framework.

Building on these research directions, the present project develops a predictive and decision-support framework for digital advertising placement optimization. Using contextual advertising data, a machine learning model is trained to predict expected conversions per impression based on demographic, contextual, and advertisement features. The predictive model is then integrated into a contextual decision policy that selects the advertisement placement predicted to maximise expected conversion outcomes. To evaluate the practical value of this approach, the model-based placement strategy is compared with several baseline strategies, including random placement selection, most frequent placement selection, and global best placement selection. An oracle benchmark is also used to represent the theoretical upper bound of achievable performance. Through this framework, the project investigates whether predictive modelling can meaningfully improve advertising placement decisions in a contextual decision environment.

3. Work Undertaken

a. 3.1 Analytical System Overview

This project implemented a complete analytical pipeline designed to analyse digital advertising performance and support data-driven advertisement placement decisions. The system integrates multiple stages of data processing, feature engineering, exploratory analysis, predictive modelling, and decision policy evaluation.

The workflow begins with loading and preprocessing a structured dataset containing demographic, contextual, and advertisement attributes. After cleaning and standardising the data, additional temporal and contextual features are generated to capture patterns that may influence advertising behaviour. A new performance metric, expected conversions per

impression (ECPI), is then constructed to represent the overall effectiveness of an advertisement impression.

Following preprocessing, a series of exploratory visualisation procedures are implemented to analyse relationships between contextual variables and advertising performance metrics. These visualisations are produced through reusable analytical functions that generate distribution plots, category comparisons, correlation heatmaps, and feature-effect graphs.

The analytical pipeline then proceeds to predictive modelling. A machine learning model based on gradient boosting regression is trained to estimate expected conversions per impression using contextual and advertisement features. The trained model is integrated into a contextual decision framework that evaluates alternative advertisement placement strategies.

To assess the effectiveness of model-driven placement decisions, a policy evaluation framework compares the model-based recommendation strategy with several baseline approaches, including random placement selection, global best placement selection, and most frequent placement selection. An oracle benchmark is also included to represent the theoretical upper bound of achievable performance.

Finally, the analytical components of the system are integrated into an interactive dashboard developed using Streamlit. This dashboard allows users to explore advertisement performance across different contexts and visualise analytical results through interactive charts.

b. 3.2 Dataset Description

The analysis begins by loading the advertising dataset using the Python Pandas library. The dataset is stored in a CSV file and contains records representing advertisement impressions and associated contextual information.

The dataset includes demographic variables describing user characteristics, contextual variables describing environmental conditions, and advertisement attributes describing the advertisement itself. The primary variables included in the dataset are summarised below.

Variable	**Description**
Age	Age of the user exposed to the advertisement
Income	Estimated income level of the user
Ad Topic	Topic category of the advertisement
Ad Type	Advertisement format
Placement	Platform where the advertisement appears
Location	Geographic context of the user

Timestamp	Time at which the advertisement was displayed
Click Through Rate	Probability of a click occurring
Conversation Rate	Probability of conversion following a click

An initial inspection of the first rows of the dataset is performed to verify that values are correctly loaded and to understand the basic structure of the data. This step ensures that variables are properly formatted and confirms that the dataset contains the necessary fields required for constructing the ECPI performance metric and training predictive models.

Performing this inspection early in the pipeline helps identify potential inconsistencies in column naming or data types before further preprocessing operations are applied.

c. 3.3 Column Detection and Standardisation

To ensure robustness of the analytical pipeline, a column detection mechanism was implemented to automatically identify key variables within the dataset. In real-world datasets, column names may vary slightly depending on the data source or formatting conventions. For example, click-through rate may appear as “CTR”, “click_through_rate”, or other similar variations.

To address this issue, custom functions were implemented to search for candidate column names using pattern matching. The functions find_col() and find_col_contains() scan the dataset’s column names and match them with predefined candidate lists representing expected variables.

This automated detection process identifies the columns corresponding to:

Variable	**Description**
CTR	Click-through rate, representing the proportion of impressions that result in a click
Conversion Rate	Proportion of clicks that result in a successful conversion
Clicks	Number of user clicks recorded for an advertisement
Timestamp (Click Time)	Time at which the user interaction occurred, used to extract temporal features such as hour, day, and weekend indicators
Ad Placement	Platform or channel where the advertisement is displayed (e.g. social media, website,

	search engine)
Gender	User demographic variable indicating gender category
Location	User geographic classification (e.g. urban, suburban, rural)

Once detected, these columns are stored as variables within the analytical pipeline to ensure consistent referencing throughout the preprocessing and modelling stages. This approach improves the flexibility of the system and reduces the risk of errors caused by inconsistent naming conventions.

d. 3.4 Data Cleaning and Pre-Processing

After identifying the relevant columns, several preprocessing steps are applied to prepare the dataset for analysis and modelling.

First, categorical variables stored as object types are converted to string format and trimmed to remove leading or trailing whitespace. This step ensures that categorical labels remain consistent and prevents duplicate categories caused by formatting inconsistencies. Variables are then categorised according to their measurement type. Continuous variables, including age, income, click-through rate, conversion rate, and ECPI, are treated as numerical features, while discrete categorical variables such as advertisement placement, topic, location, and gender are processed as categorical inputs. This distinction ensures that appropriate preprocessing techniques are applied to each variable type during model training.

Next, numeric variables including CTR, conversion rate, and clicks are converted to numeric data types using coercion. This conversion ensures that non-numeric entries are handled as missing values rather than causing computational errors. The timestamp variable representing click time is converted into a datetime format, enabling the extraction of temporal features used in subsequent analysis. Rows containing missing values for CTR or conversion rate are removed from the dataset, as these variables are required to compute the ECPI metric. After these steps, the dataset is standardised and prepared for feature engineering and modelling tasks.

e. 3.5 Temporal Feature Engineering

To capture time-dependent patterns in advertising behaviour, a set of temporal features is derived from the timestamp variable representing click time.

Using Pandas datetime functionality, the timestamp is decomposed into several components, including hour of day, day of week, month, and a binary indicator for weekend activity. These features provide a structured representation of temporal context, enabling the analysis to account for behavioural variation across different time periods.

In particular, hour of day captures intra-day fluctuations in user activity, while day of week and weekend indicators reflect differences between weekday and weekend behaviour. The inclusion of month allows for the identification of broader seasonal patterns. Together, these features allow the model to capture temporal dynamics that may influence both engagement and conversion behaviour.

Once the temporal features are extracted, the original timestamp column is removed to avoid redundancy while retaining the relevant information in a more structured and model-compatible form.

f. 3.6 Construction of the Target Metric

A central component of this project is the construction of a unified performance metric representing the expected effectiveness of an advertisement impression. While click-through rate and conversion rate are commonly used in digital advertising analytics, they capture different stages of the user interaction process. Click-through rate measures the probability that a user clicks on an advertisement after seeing it, while conversion rate measures the probability that a click leads to a successful action such as a purchase or registration.

To represent the complete advertising outcome chain, a new metric called Expected Conversions per Impression (ECPI) is constructed. This metric is defined as the product of click-through rate and conversion rate:

ECPI = CTR × Conversion Rate

The ECPI metric represents the expected probability that a single advertisement impression results in a successful conversion. By combining engagement probability with post-click success probability, this metric provides a more comprehensive measure of advertisement effectiveness than analysing CTR or conversion rate independently. Within the analytical pipeline, ECPI is calculated for each observation and stored as the target variable used in subsequent exploratory analysis and predictive modelling stages.

g. 3.7 Exploratory Visualisation Framework

To support systematic analysis of advertisement performance across contextual variables, a reusable visualisation framework was developed to enable consistent comparison of ECPI across demographic, contextual, and advertisement-related features. Implemented using Plotly Express, the framework facilitates interactive exploration while maintaining a standardised structure for analysing category-level performance. In particular, performance is evaluated by aggregating ECPI across categorical groupings, allowing differences between variables such as placement channel, topic, advertisement type, geographic location, demographic attributes, and temporal segments to be examined in a comparable manner. To ensure the reliability of these comparisons, categories with insufficient observations are excluded through a minimum sample threshold, reducing the risk of misleading averages driven by limited data. This approach ensures that performance differences are assessed consistently across variables, enabling clearer identification of patterns while maintaining robustness in the analysis.

h. 3.8 Feature Effect Analysis Functions

In addition to category-level comparisons, the analytical pipeline incorporates a general feature-effect analysis framework designed to examine how different variables influence key performance metrics. This framework enables the systematic evaluation of relationships between predictor variables and outcomes such as click-through rate, conversion rate, and ECPI, adapting the analysis based on the structure of each variable. For categorical features, performance is assessed through aggregation across categories, allowing direct comparison of average outcomes, while continuous variables are analysed through discretisation into quantile-based intervals to preserve meaningful variation across value ranges. This approach ensures that both discrete and continuous features can be evaluated within a consistent analytical structure, enabling a comprehensive assessment of how individual variables relate to performance. By automating this process, the framework supports efficient exploration of feature-performance relationships across multiple variables and target metrics, reducing manual intervention while maintaining consistency and interpretability in the analysis.

i. 3.9 Correlation Analysis Implementation

To examine relationships between numerical variables, a correlation analysis was conducted to identify potential dependencies relevant to advertising performance. A correlation matrix was computed using the Pearson coefficient across demographic, behavioural, temporal, and outcome variables, including ECPI. The results were visualised using a heatmap to provide a structured overview of the strength and direction of pairwise relationships, enabling rapid identification of variables that exhibit meaningful associations. This analysis supports the modelling process by highlighting potential feature interactions and dependencies within the dataset, providing insight into the underlying numerical structure prior to predictive modelling.

j. 3.10 Advertisement Placement Mapping

To examine how advertisement placement channels influence performance, a visualisation framework was developed to compare click-through rate, conversion rate, and overall effectiveness across placements. The dataset is aggregated by placement channel, with mean click-through rate and mean conversion rate computed for each category. These aggregated values are represented in a two-dimensional scatter plot, enabling direct comparison of engagement and conversion performance across platforms.

To incorporate data reliability into the analysis, marker size is scaled according to the number of observations within each placement category. This allows the visualisation to distinguish between consistently observed patterns and those based on limited data, providing a more robust interpretation of performance differences.

The analysis is further extended through a three-dimensional representation, where click-through rate, conversion rate, and observation frequency are visualised simultaneously, with colour intensity used to reflect ECPI. This multi-dimensional mapping enables the identification of trade-offs between engagement, conversion efficiency, and data volume across placement channels. Together, these visualisations provide a comprehensive view of placement

performance, allowing patterns and relative differences to be interpreted across multiple dimensions rather than through isolated metrics.

k. 3.11 Predictive Modelling Pipeline

Following exploratory analysis, a predictive modelling pipeline was developed to estimate expected conversions per impression (ECPI) using contextual and advertisement-related features. The model is implemented using the HistGradientBoostingRegressor from the Scikit-learn library, which is well suited to structured tabular data due to its ability to capture non-linear relationships and interaction effects between heterogeneous variables. To enable robust evaluation, the dataset is partitioned into training and testing subsets using an 80–20 split, ensuring that model performance reflects generalisation to unseen data rather than in-sample fit.

A preprocessing pipeline is constructed using Scikit-learn's ColumnTransformer and Pipeline utilities, allowing transformations to be applied systematically based on variable type. Numerical features are processed using median imputation to handle missing values, while categorical features are processed using most-frequent imputation followed by one-hot encoding, enabling categorical information to be incorporated into the model. To prevent target leakage, variables directly used in the construction of ECPI, including click-through rate and conversion rate, are excluded from the feature set, ensuring that the model learns from contextual signals rather than reconstructing the target variable. The preprocessing and modelling steps are integrated into a single pipeline, improving reproducibility and ensuring consistent application of transformations during both training and evaluation.

l. 3.12 Context Generation and Feature Grouping

In order to support contextual decision policies, the analytical pipeline constructs contextual groupings that represent distinct advertising environments. Rather than treating each observation independently, demographic and temporal variables are transformed into higher-level categories that reflect meaningful behavioural segments. Demographic variables such as age and income are grouped using binning techniques, with age divided into broader age groups and income partitioned into quantile-based categories, while temporal variables are structured into contextual categories by mapping hour values to time-of-day segments such as night, morning, afternoon, and evening, and representing day-of-week values as categorical labels. These transformations reduce noise in continuous variables while preserving interpretable patterns in user behaviour, allowing observations to be organised into comparable contexts. As a result, advertisement performance can be evaluated within consistent environmental conditions, enabling the system to identify how outcomes vary across demographic and temporal states and supporting more informed and adaptive placement decisions.

m. 3.13 Recommendation Policy Implementation

Using the trained predictive model, a contextual recommendation policy was implemented to determine the optimal advertisement placement for a given context. For each contextual

scenario, the system evaluates a set of candidate placement options and uses the model to estimate the expected ECPI associated with each alternative. The placement corresponding to the highest predicted ECPI is then selected, effectively operationalising the model within an optimisation framework. This approach transforms the predictive model from a purely analytical tool into a decision-support mechanism, where placement selection is guided by expected outcomes conditional on the observed context. As a result, the policy enables adaptive allocation decisions that respond to variation in demographic and temporal conditions, rather than relying on static or aggregate strategies.

n. 3.14 Policy Evaluation Framework

To evaluate the effectiveness of the model-based placement recommendations, a policy evaluation framework was implemented to compare the performance of alternative decision strategies under consistent conditions. The framework includes a range of policies with varying levels of information and complexity, including a random policy that selects placements without contextual awareness, a most frequent policy based on historical prevalence, and a global best policy that selects the placement with the highest average ECPI across the dataset. These are compared against the model-based policy, which selects the placement predicted to maximise ECPI for each context, as well as an oracle benchmark representing the best possible decision based on observed outcomes. Each policy is evaluated across multiple contextual scenarios, with expected ECPI calculated and summarised to enable direct comparison of decision effectiveness. This framework provides a systematic basis for assessing whether predictive modelling leads to meaningful improvements over simpler heuristic approaches, while also establishing an upper bound on achievable performance.

o. 3.15 Prediction Validation

To assess the predictive performance of the trained model, predicted ECPI values are compared with observed values from the test dataset using a scatter plot representation. Each point corresponds to an individual observation, allowing the relationship between predicted and actual outcomes to be examined directly, with a reference line indicating perfect prediction. The degree to which observations align with this reference provides a visual indication of model accuracy, where closer alignment reflects stronger predictive performance. This form of validation complements numerical evaluation metrics by revealing the distribution of prediction errors and highlighting any systematic deviations or bias in the model. As such, it provides a more detailed assessment of how effectively the model captures underlying patterns in advertisement performance beyond aggregate performance measures.

p. 3.16 Interactive Dashboard Implementation

To enhance the interpretability and practical usability of the analytical outputs, an interactive dashboard was developed using the Streamlit framework. The dashboard provides a user interface that enables dynamic exploration of advertisement performance across contextual variables, including placement, topic, demographic attributes, and temporal features. Through interactive visualisations such as ECPI distributions, category comparisons, correlation heatmaps, and temporal performance charts, users can examine how performance varies under different conditions without direct interaction with the underlying code. This interface bridges the

gap between predictive modelling and practical decision-making by transforming model outputs into accessible and interpretable insights. As a result, the system supports exploratory analysis and facilitates more informed, context-aware decisions in advertising placement.

4. Outputs

a. Descriptive Statistics and Baseline Performance

The first step in evaluating the analytical pipeline is to examine the distribution of the ECPI metric produced during preprocessing. This establishes a baseline for advertising performance and indicates how outcomes are distributed across impressions.

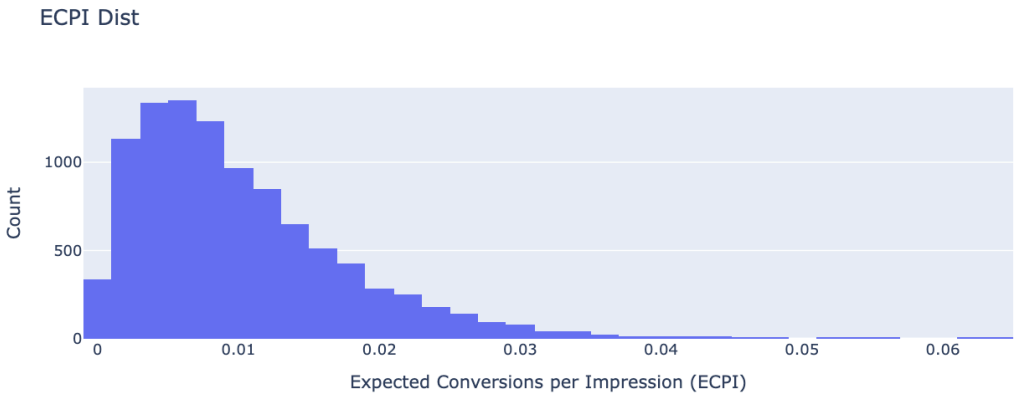


Figure 1. Distribution of Expected Conversions per Impression (ECPI).

The distribution is strongly right-skewed, with most observations concentrated at low ECPI values and a small number of impressions generating higher outcomes. This indicates that high-performing advertisement exposures are relatively rare, while the majority contribute minimal expected value. This pattern suggests that improvements in overall performance depend on accurately identifying and targeting a limited subset of high-value contexts rather than optimising across all impressions uniformly. It also indicates that average performance measures may mask underlying variability, as a small proportion of observations contributes disproportionately to total expected conversions.

b. 4.2 Advertisement Placement Performance

Advertisement placement represents a key contextual variable influencing digital advertising outcomes, as different environments expose users to advertisements under varying levels of engagement and attention. To examine this relationship, the mean expected conversions per impression (ECPI) was calculated for each placement category, providing a measure of overall effectiveness.

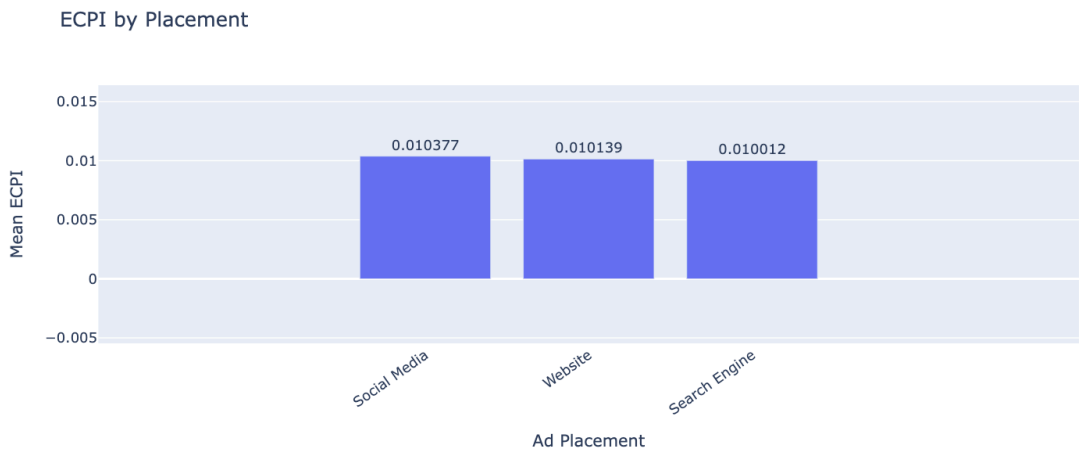


Figure 2. Mean Expected Conversions per Impression by Advertisement Placement.

Social media placements exhibit the highest mean ECPI (≈0.01038), followed by websites (≈0.01014) and search engine placements (≈0.01001). While this indicates that social media environments perform marginally better, the differences across placement categories remain small in magnitude.

This limited variation suggests that placement alone does not strongly determine advertising performance within the dataset. Although social media may benefit from higher user engagement and more effective targeting mechanisms, the narrow spread in ECPI values indicates that these advantages do not translate into substantial differences in expected outcomes.

From a modelling perspective, this implies that placement is likely a weak standalone predictor of ECPI and that stronger predictive performance will depend on interactions between multiple contextual variables. Consequently, optimisation strategies based solely on placement selection are unlikely to yield significant improvements without incorporating additional features.

c. 4.3 Advertisement Topic Performance

Advertisement topic represents another contextual variable that may influence user engagement and conversion behaviour, as different content categories can appeal to users with varying interests and purchase intent. To evaluate this effect, the mean expected conversions per impression (ECPI) was calculated for each advertisement topic category.

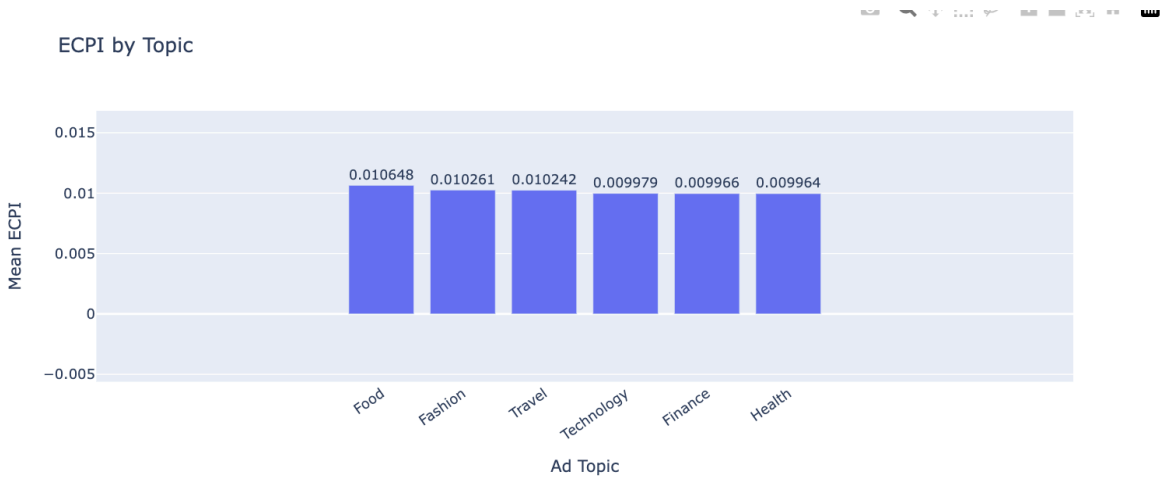


Figure 3. Mean Expected Conversions per Impression by Advertisement Topic.

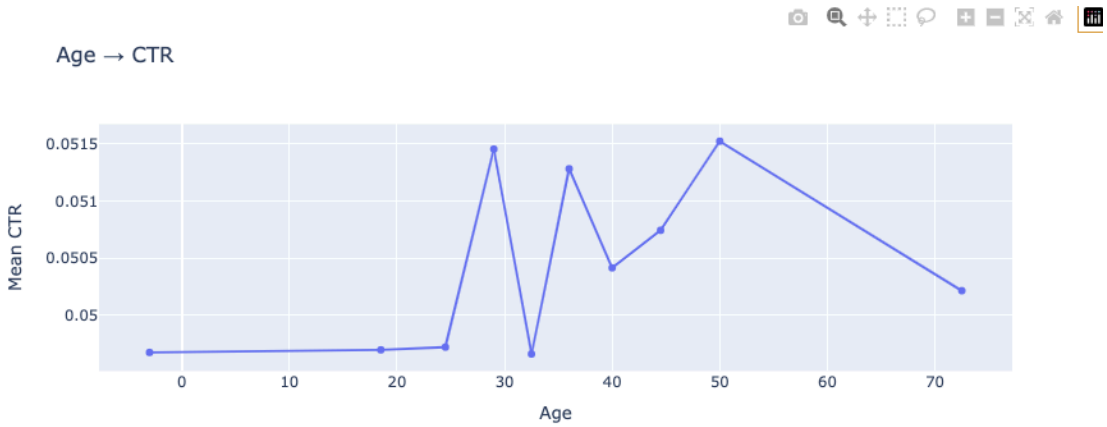
The results show modest variation in ECPI across topic categories, with certain categories such as food achieving slightly higher values relative to others. However, the overall spread in ECPI remains narrow, indicating limited differentiation in performance between topics.

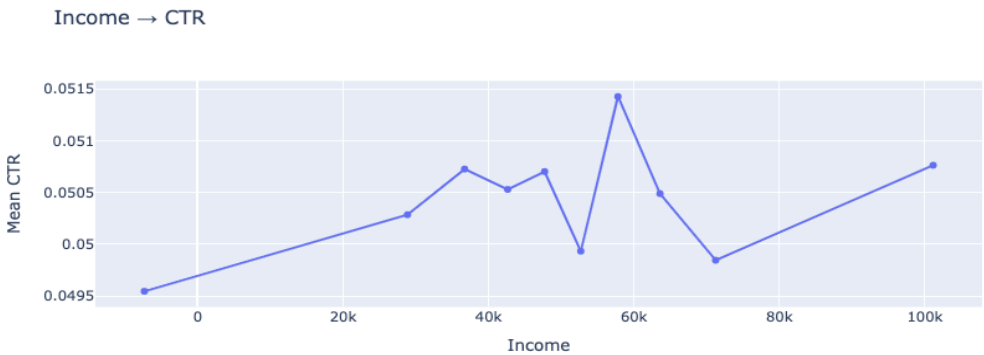
This suggests that advertisement topic alone has a relatively weak influence on expected conversion outcomes within the dataset. While some categories may generate marginally higher engagement due to relevance or consumer interest, these effects are not sufficiently large to produce strong performance differences.

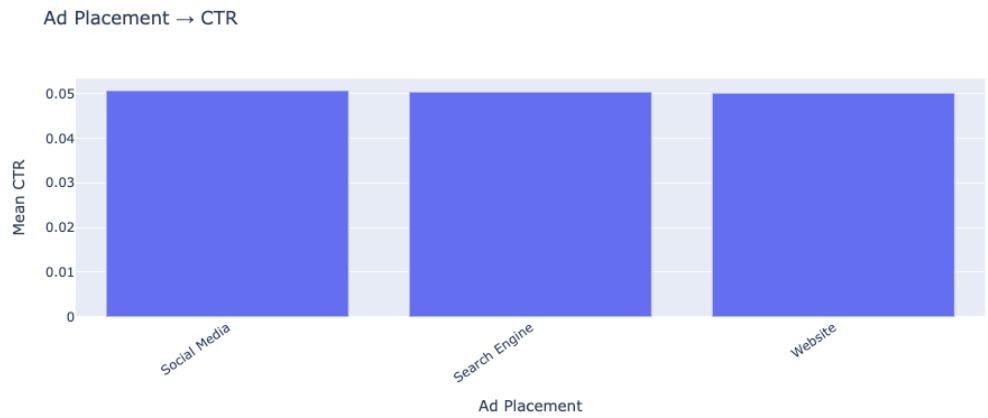
From a modelling perspective, this implies that topic is unlikely to act as a strong independent predictor of ECPI. Instead, its contribution is likely to emerge through interactions with other contextual variables such as user demographics and timing. As a result, optimisation strategies based solely on topic selection are expected to have limited impact unless combined with additional contextual information.

d. 4.4 Drivers of Click-Through Rate

To further examine how individual contextual variables influence advertising outcomes, feature–effect plots were used to analyse the relationship between selected variables and observed click-through rate (CTR). Figure 4 presents these relationships across key variables, including age, income, advertisement placement, and advertisement topic.


Age → CTR
Mean CTR
0.0515
0.051
0.0505
0.05
0
10
20
30
40
50
60
70
Age


Income → CTR
Mean CTR
0.0515
0.051
0.0505
0.05
0.0495
0
20k
40k
60k
80k
100k
Income


Ad Placement → CTR
Mean CTR
0.05
0.04
0.03
0.02
0.01
0
Social Media
Search Engine
Website
Ad Placement

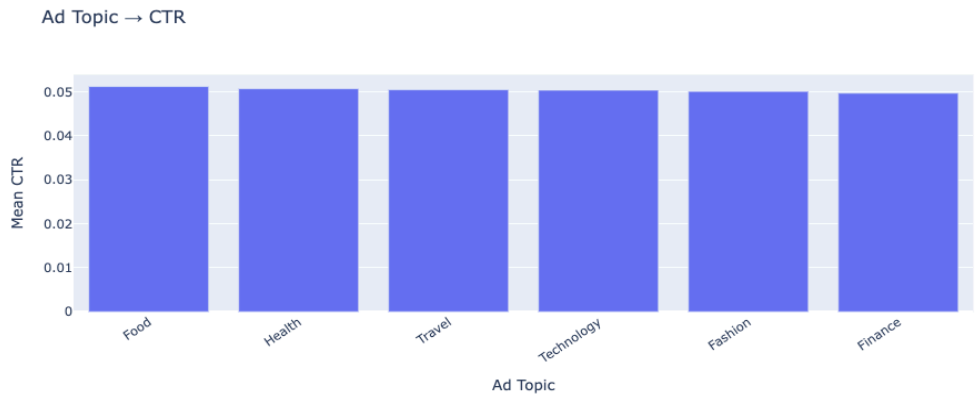


Figure 4. Feature-effect plots illustrating the relationship between contextual variables and click-through rate (CTR): (a) age, (b) income, (c) advertisement placement, and (d) advertisement topic.

The results indicate that variation in CTR across individual variables is present but limited in magnitude. Age exhibits minor fluctuations, with slightly higher click-through rates observed in specific age ranges, suggesting modest variation in user engagement across demographic groups. Similarly, income shows a weak positive relationship with CTR, where higher income categories display marginally increased engagement.

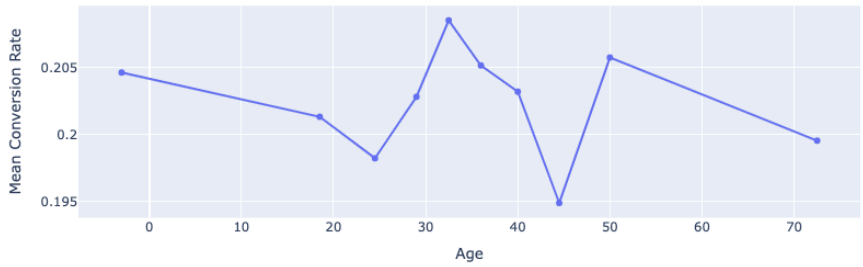
Differences across advertisement placement and topic are also observable but remain relatively small. Social media placements demonstrate slightly higher click-through rates compared to other channels, while certain topics show marginal performance advantages. However, these differences are not substantial, indicating that no single variable produces strong shifts in click behaviour.

Overall, the feature–effect analysis suggests that click behaviour is only weakly influenced by individual variables when considered independently. This reinforces the view that advertising performance is not driven by isolated factors, but rather by the interaction of multiple contextual conditions. Consequently, models that capture non-linear relationships and feature interactions are more appropriate for accurately estimating performance outcomes.

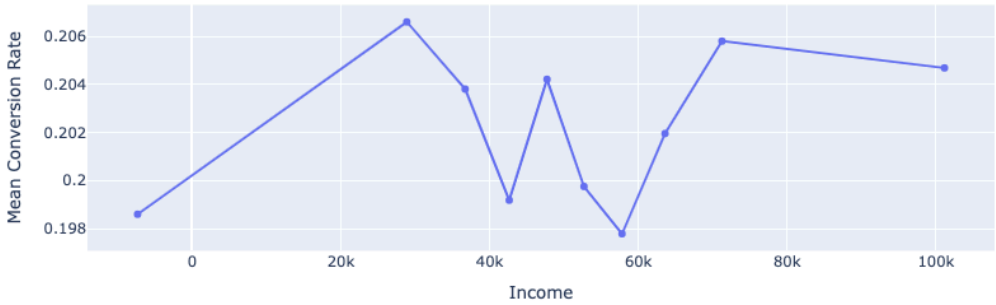
e. 4.5 Drivers of Conversion Rate

To analyse how individual contextual variables influence user behaviour, feature–effect plots were constructed to examine the relationship between selected variables and observed conversion rates.

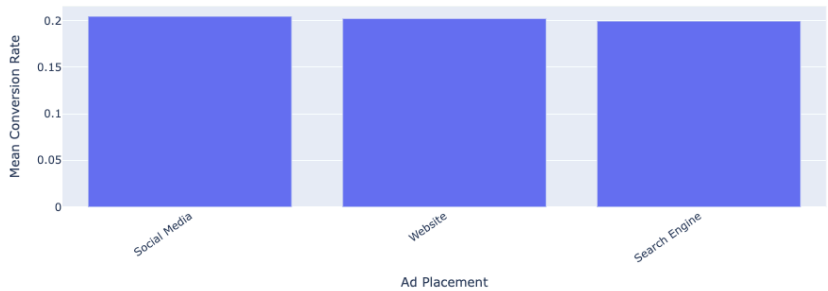
Age → Conversion Rate



Income → Conversion Rate



Ad Placement → Conversion Rate



Ad Topic → Conversion Rate

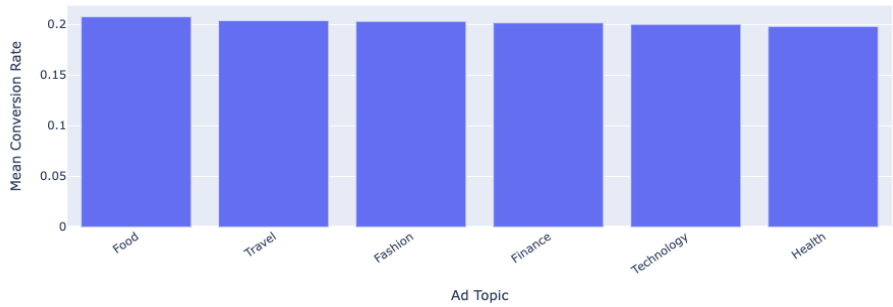


Figure 5. Feature–effect plots illustrating the relationship between contextual variables and conversion rate: (a) Age, (b) Income, (c) Advertisement Placement, and (d) Advertisement Topic.

The results indicate that variation in conversion rate across individual variables is present but limited in magnitude. Age shows minor fluctuations, with slightly higher conversion rates observed within specific ranges, suggesting some variation in user responsiveness across demographic groups. Income exhibits a weak positive relationship with conversion probability, with higher income categories demonstrating marginally higher conversion rates.

Differences across advertisement placement and topic are also observable but remain small. Social media placements show slightly higher conversion rates compared to other channels, while certain topics display marginal performance advantages. However, the overall spread across categories is narrow.

These results indicate that individual variables have limited explanatory power when considered independently. The relatively small differences across all features suggest that conversion behaviour is not driven by any single factor, but rather by the interaction of multiple contextual conditions. This supports the use of models capable of capturing non-linear relationships and feature interactions, as linear or single-variable approaches are unlikely to fully explain variation in conversion outcomes.

In addition, the relatively smooth and consistent patterns observed across variables may reflect underlying structure within the dataset, suggesting that the relationships identified may be less pronounced in real-world settings where greater noise and behavioural variability are present.

f. 4.6 Model Comparisons

To evaluate the effectiveness of different predictive approaches, multiple models were compared in their ability to estimate expected conversions per impression (ECPI). These included a baseline linear model (Ridge Regression) and two non-linear ensemble methods (Random Forest and HistGradientBoosting), allowing for a comparison between linear and more flexible modelling techniques.

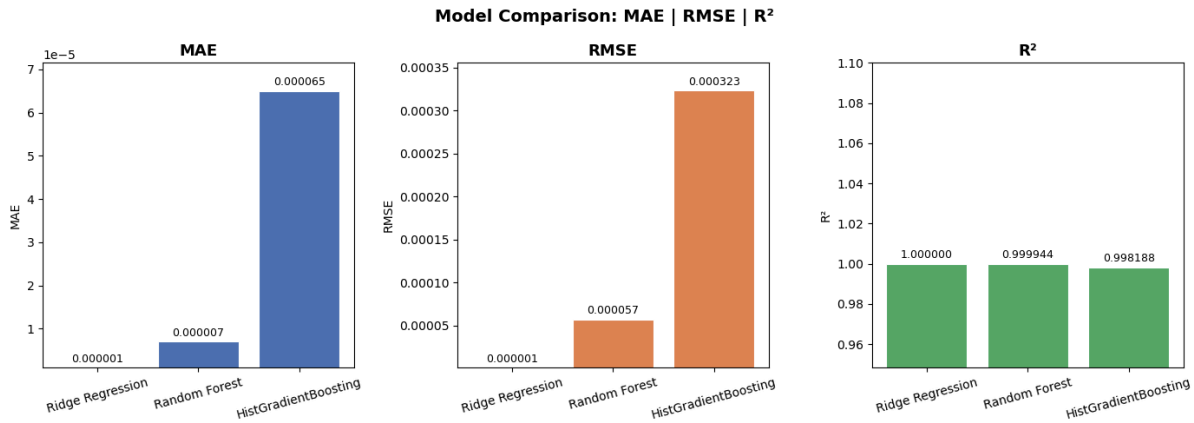


Figure 6: Model comparison across MAE, RMSE, and R^2 for Ridge Regression, Random Forest, and HistGradientBoosting.

The results indicate that both models achieve extremely high predictive performance, with the gradient boosting model marginally outperforming the linear baseline. This suggests that the underlying relationships within the dataset are highly structured and can be captured effectively even by relatively simple models. The strong performance of the linear model further implies that a large proportion of the variance in ECPI can be explained through additive relationships between features.

However, the near-perfect performance observed, particularly for the linear model, warrants careful interpretation. In real-world behavioural prediction tasks, such levels of accuracy are uncommon due to noise, unobserved variables, and dynamic user behaviour. This raises the possibility that the dataset contains highly deterministic patterns or residual structure that simplifies the prediction task. As a result, while the models appear highly effective within this controlled setting, their performance may not generalise to more complex, real-world advertising environments.

The marginal improvement achieved by the gradient boosting model suggests that non-linear interactions between variables do exist, but their contribution is limited relative to the overall structure captured by linear effects. This indicates that increasing model complexity yields diminishing returns in this context, as the majority of predictive signal is already captured by simpler relationships.

From a practical perspective, these findings highlight an important trade-off between model complexity and interpretability. While more advanced models offer slight improvements in predictive accuracy, simpler models may be sufficient for deployment when interpretability, computational efficiency, and stability are prioritised.

Overall, the comparison demonstrates that predictive performance in this task is less constrained by model choice and more influenced by the structure and quality of the underlying data. This reinforces the idea that improvements in feature design and data richness may be more impactful than further increases in model complexity.

g. 4.7 Model Performance

To estimate expected conversions per impression (ECPI), a predictive model was implemented using a HistGradientBoostingRegressor. Model performance was evaluated on a held-out test set using standard regression metrics. The model achieves a very high level of predictive accuracy, with predicted ECPI values closely aligned to observed outcomes. Both the mean absolute error and root mean squared error are extremely low, while the coefficient of determination indicates that the model explains almost all variance in the target variable. This suggests that the dominant patterns within the dataset are captured effectively by the selected features and modelling approach.

However, the magnitude of this performance warrants careful interpretation. In behavioural prediction tasks, such near-perfect explanatory power is uncommon due to noise, unobserved heterogeneity, and dynamic user behaviour. The results therefore indicate that the dataset contains highly structured relationships, which simplify the prediction task and reduce uncertainty in the target variable. In particular, the construction of ECPI as a derived metric may embed systematic relationships that are easier for the model to learn.

From a modelling perspective, these results confirm the suitability of gradient boosting for structured tabular data. The model is able to capture non-linear relationships and interaction effects between contextual variables without requiring extensive manual feature design. This aligns with earlier findings that individual variables exhibit weak effects in isolation but become more informative when combined.

At the same time, the strength of the observed performance highlights a limitation in terms of generalisability. While the model demonstrates strong internal validity within the dataset, its external validity is likely constrained. Real-world advertising environments typically exhibit greater variability, noise, and temporal instability, which would reduce predictive performance. As a result, the reported accuracy should be interpreted as an upper bound achievable under controlled data conditions rather than a direct reflection of real-world performance.

Overall, the model performance demonstrates that ECPI can be predicted with high precision within the given dataset. However, the results also emphasise that predictive accuracy is influenced not only by model capability, but by the structure and signal strength of the data. This reinforces the importance of evaluating models within realistic contexts when assessing their practical effectiveness.

h. 4.8 Advertisement Allocation Results

To evaluate the practical effectiveness of the predictive modelling framework, an advertisement allocation strategy was implemented using predicted expected conversions per impression (ECPI). The objective was to assess whether model-based decision-making improves advertising outcomes relative to alternative allocation strategies.

Several allocation strategies were evaluated across identical contexts. The model-based policy selects the advertisement placement that maximises predicted ECPI for each context. This approach was compared against benchmark strategies, including a most frequent policy based on historical prevalence, a global best policy selecting the overall highest-performing placement, a random allocation policy, and an oracle policy representing the optimal selection based on true observed outcomes.

The performance of each strategy is summarised in **Table X**, which reports both mean and median ECPI across all evaluated contexts.

Policy	**Mean Value**	**Median Value**
Model	0.010521	0.010767
Most Frequent	0.009906	0.008958
Global Best	0.009709	0.008738
Random	0.010386	0.009755
Oracle	0.013148	0.012550

As shown in **Table X**, the model-based policy achieves the highest performance among all feasible strategies, with a mean ECPI of 0.010521 and a median of 0.010767. This represents an improvement over the random policy (mean = 0.010386; median = 0.009755), as well as more substantial gains relative to the most frequent (mean = 0.009906) and global best strategies (mean = 0.009709). These results indicate that incorporating predictive modelling into allocation decisions leads to consistently higher expected outcomes.

The magnitude of improvement, however, remains relatively small. The model-based policy improves mean performance by approximately 1.3% over random allocation and by 6–8% over static baseline strategies. As reflected in Table X, the relatively close values across all policies suggest that variation in ECPI across placement options is limited. This indicates that, in many contexts, alternative placements yield similar expected outcomes, reducing the potential gains from optimisation.

The comparatively strong performance of the random policy further supports this interpretation. Despite lacking any contextual awareness, random allocation achieves a mean ECPI close to that of the model-based policy. This suggests that the underlying differences between placements are modest, and that only incremental improvements can be achieved through more informed decision-making.

In contrast, the most frequent and global best strategies perform notably worse, as shown in Table X. These approaches rely on aggregate statistics and fail to account for context-specific variation in user behaviour and advertisement conditions. As a result, they are unable to adapt

to differences across contexts and consistently underperform relative to both the model-based and random policies.

The oracle policy achieves the highest performance overall, with a mean ECPI of 0.013148 and a median of 0.012550, as reported in Table X. This represents the theoretical upper bound of achievable performance. The gap between the oracle and the model-based policy is substantial, at approximately 25%, indicating that while the model captures meaningful patterns, it does not fully exploit all available information. This gap reflects both modelling limitations and the inherent unpredictability of user behaviour.

Importantly, the combination of a strong random baseline and a large oracle gap suggests that the dataset contains limited observable variation between actions but retains latent structure that is not fully captured by the model. This indicates diminishing returns to optimisation under the current feature set, where improvements are constrained by weak signal strength in the available variables.

Overall, the results presented in Table X demonstrate that integrating predictive modelling with contextual allocation policies leads to more effective utilisation of advertising opportunities. While the improvements are incremental, they are consistent and achieved under realistic conditions. In large-scale digital advertising systems, even small increases in expected value can translate into significant cumulative gains, supporting the practical relevance of the proposed approach.

i. 4.9 Interactive Dashboard

To enhance the interpretability and practical applicability of the analytical pipeline, an interactive dashboard was developed using Streamlit and Plotly. The dashboard provides a user-facing interface that allows dynamic exploration of advertisement performance across multiple contextual dimensions, bridging the gap between predictive modelling outputs and actionable decision-making.

The dashboard integrates the full analytical pipeline, including data preprocessing, feature engineering, and model outputs, into a unified environment. Users are able to upload datasets, apply contextual filters, and visualise performance metrics in real time. This enables flexible analysis of expected conversions per impression (ECPI) across different combinations of demographic, temporal, and advertisement-related variables.

A key feature of the interface is the dynamic filtering system, which allows users to subset the dataset based on variables such as advertisement placement, advertisement type, topic, location, gender, weekday, and month. This functionality enables context-specific analysis, allowing users to examine how performance varies across different conditions and to identify patterns that may not be visible in aggregate statistics.

The dashboard presents multiple visual outputs designed to support analytical interpretation. Distribution plots illustrate the spread and variability of ECPI, providing insight into the consistency of advertisement performance. Bar charts summarise average ECPI across selected categorical variables, enabling direct comparison between different placements or demographic groups. In addition, correlation heatmaps are used to visualise relationships between numerical variables, supporting the identification of potential dependencies within the dataset.

The inclusion of summary metrics, such as mean ECPI, median ECPI, and the number of unique advertisement placements, provides users with immediate quantitative feedback based on selected filters. This allows for rapid assessment of performance under different contextual scenarios without requiring manual recalculation or external analysis tools.

From a decision-making perspective, the dashboard operationalises the predictive modelling framework by allowing users to interactively explore which contextual combinations lead to higher expected conversion outcomes. Rather than relying solely on static results, the interface enables exploratory analysis, supporting more informed and adaptive allocation decisions.

However, the functionality of the dashboard is influenced by the characteristics of the underlying dataset. As observed in earlier sections, variation in ECPI across contexts is relatively limited. This constrains the visual differentiation between categories and may reduce the apparent impact of filtering operations. As a result, while the dashboard remains effective for exploratory analysis, its ability to reveal strong patterns is dependent on the variability and richness of the input data.

In addition, the dashboard currently operates as an analytical tool rather than a fully automated decision system. While it enables users to identify high-performing contexts, it does not directly implement real-time allocation decisions or integrate with live advertising platforms. Extending the system to include automated recommendation outputs or real-time data integration would further enhance its practical applicability.

Overall, the interactive dashboard provides a critical link between predictive modelling and user interpretation. By transforming model outputs into accessible visual insights, it supports data-driven decision-making and improves the usability of the analytical framework. This demonstrates the importance of combining technical modelling approaches with effective visualisation tools to ensure that analytical results can be meaningfully applied in real-world contexts.

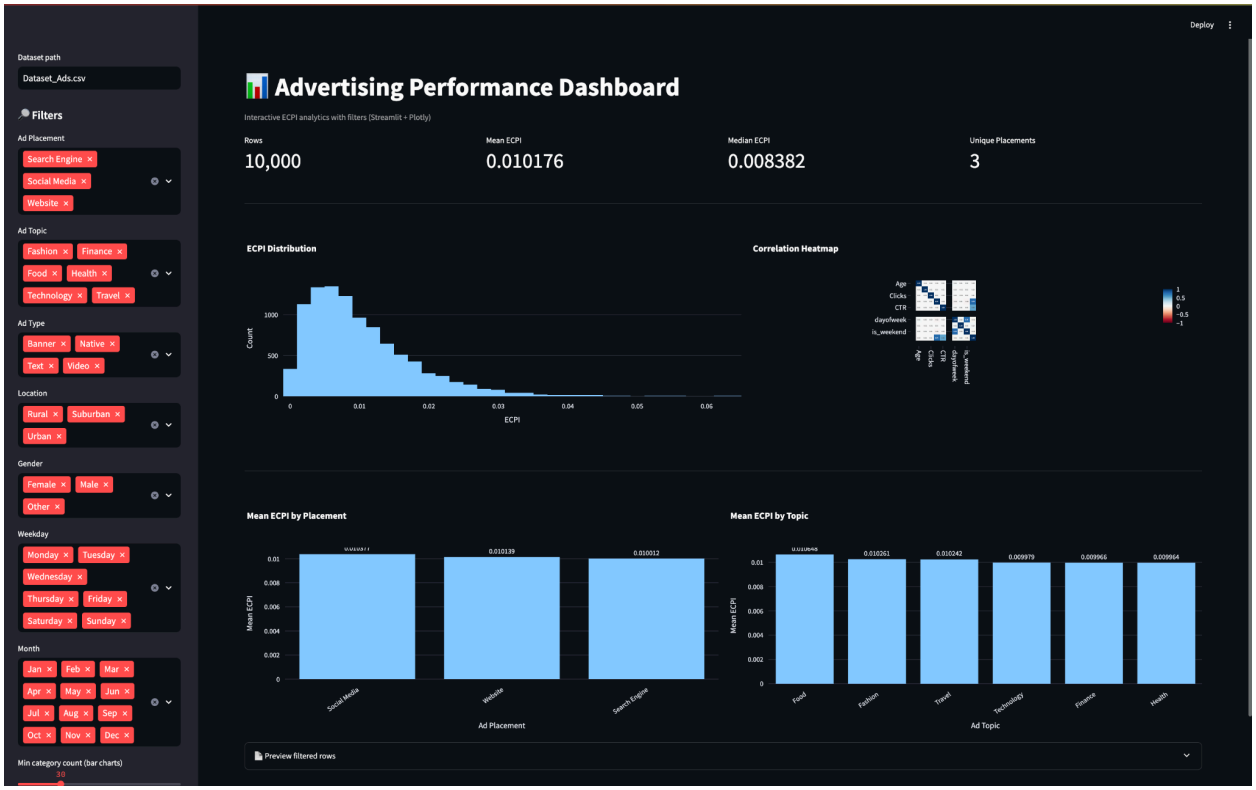


Figure 7. Interactive advertising performance dashboard showing summary indicators, ECPI distribution, correlation heatmap, and contextual performance charts.

j. 4.10 Discussion

The findings of this investigation indicate that differences in ECPI across advertisement placements and topics are present but limited in magnitude, suggesting that these variables alone are weak drivers of advertising performance. While social media placements and certain content categories demonstrate marginally higher ECPI values, the overall spread across categories remains narrow. This implies that optimisation strategies based solely on single variables such as placement or topic are unlikely to produce substantial improvements, as performance appears to be only weakly separable along these dimensions.

In contrast, the predictive modelling results demonstrate that ECPI can be estimated with a high degree of accuracy when multiple contextual variables are considered simultaneously. This highlights that advertising performance is better understood as a function of interacting factors rather than isolated features, with demographic and temporal variables contributing jointly to more precise predictions. However, the exceptionally high predictive performance observed suggests that the dataset contains highly structured relationships, which may simplify the modelling task. In real-world advertising environments, where user behaviour is more variable and influenced by unobserved factors, predictive accuracy would likely be lower and performance differences more pronounced.

Several limitations should therefore be considered. The use of aggregated metrics such as mean ECPI provides a clear summary of performance but may obscure substantial variation within categories, limiting the identification of more granular optimisation opportunities. In addition, the feature set is restricted to a subset of contextual variables, excluding potentially influential factors such as creative design, targeting strategy, and platform-specific mechanisms. This constrains the extent to which the model can fully represent real-world advertising dynamics. More broadly, the controlled structure of the dataset limits the generalisability of the findings, meaning that the results should be interpreted as indicative rather than definitive.

Overall, the analysis demonstrates that while predictive modelling improves the understanding and optimisation of advertising performance, its effectiveness is fundamentally constrained by the underlying structure and signal strength of the data. This suggests that, in practical applications, improvements in data richness and feature representation may be more impactful than further increases in model complexity. The integration of predictive modelling with contextual decision policies and interactive visualisation nevertheless provides a coherent framework for analysing and supporting advertisement placement decisions in a data-driven environment.

5. Further Work

While this investigation provides a structured analysis of advertising performance using ECPI and predictive modelling, several opportunities exist to extend and strengthen the work. A primary direction is the application of the framework to real-world datasets. The current dataset exhibits relatively stable and structured patterns, which support strong predictive performance but may not reflect the variability, noise, and external influences present in real digital advertising environments. Applying the methodology to real-world data would enable a more rigorous assessment of model generalisability and provide clearer insight into how predictive accuracy translates into decision-making performance under less controlled conditions.

A second area for development is the expansion of the feature space and modelling approach. The current model focuses on contextual and demographic variables, which only partially capture the complexity of advertising systems. Incorporating additional factors such as creative design, bidding strategies, audience targeting precision, and platform-level optimisation mechanisms would likely improve predictive performance and provide a more comprehensive understanding of the drivers of conversion outcomes. In parallel, evaluating alternative modelling techniques, including ensemble methods and more complex architectures, would allow for a more explicit examination of the trade-off between predictive accuracy, model complexity, and interpretability.

Further improvements can be made in both evaluation and system design. More rigorous validation strategies, such as cross-validation or testing on independent datasets, would strengthen confidence in model robustness, particularly in environments where data distributions evolve over time. The interactive dashboard could also be extended into a more advanced decision-support system through real-time data integration, automated placement recommendations, and scenario-based analysis tools. Finally, future work could move beyond

prediction towards direct optimisation, where advertisement allocation is explicitly modelled to maximise expected outcomes. This would establish a stronger connection between predictive modelling and operational decision-making, enhancing the practical relevance of the framework.

6. Conclusion

This investigation set out to analyse advertising performance through the lens of expected conversions per impression (ECPI) and to determine the extent to which contextual variables influence efficiency. The findings show that while individual factors such as advertisement placement and topic exhibit some variation in ECPI, these differences are relatively small when considered in isolation. This suggests that single-variable strategies are unlikely to produce significant improvements in advertising outcomes on their own.

In contrast, the predictive modelling approach demonstrates that ECPI can be estimated with a high degree of accuracy when multiple contextual variables are considered together. This confirms that advertising performance is shaped by the interaction of several factors rather than any one variable in isolation. However, while the model achieves near-perfect predictive accuracy, the corresponding improvements in allocation performance remain modest. This indicates that strong predictive performance does not necessarily translate into large practical gains, particularly in environments where variation between decision options is limited.

The results of the allocation analysis further highlight this constraint. Although the model-based policy consistently outperforms baseline strategies, the magnitude of improvement is relatively small, reflecting limited variation in ECPI across placements. At the same time, the substantial gap between the model and the oracle benchmark demonstrates that additional predictive capacity remains unexploited. Together, these findings suggest that optimisation in this context is both possible and constrained, with improvements dependent on the strength of available signals within the data.

The development of the interactive dashboard further strengthens the practical relevance of the investigation by translating analytical outputs into a format that can be explored dynamically. This enables users to examine performance across contextual conditions and supports data-driven decision-making. However, the dashboard functions primarily as a decision-support tool rather than an automated decision system, reinforcing the importance of interpretation in applied settings.

Overall, the investigation demonstrates that predictive analytics provides a more comprehensive understanding of advertising efficiency than descriptive analysis alone. To a moderate extent, predictive modelling combined with contextual decision policies improves advertising placement outcomes, but the magnitude of this improvement is constrained by limited variability in the underlying data. While the results are subject to limitations related to dataset structure and variable selection, the approach provides a strong foundation for analysing and optimising digital advertising performance in more complex and realistic environments.

7. References

Adomavicius, G. and Tuzhilin, A. (2015) 'Context-aware recommender systems', IEEE Transactions on Knowledge and Data Engineering, 23(1), pp. 1–17. Available at: https://doi.org/10.1109/TKDE.2011.218

Chaffey, D. and Ellis-Chadwick, F. (2019) Digital Marketing. 7th edn. Harlow: Pearson Education. Available at: https://www.pearson.com/en-gb/subject-catalog/p/digital-marketing/P200000000480

Friedman, J.H. (2001) 'Greedy function approximation: A gradient boosting machine', Annals of Statistics, 29(5), pp. 1189–1232. Available at: https://doi.org/10.1214/aos/1013203451

Richardson, M., Dominowska, E. and Ragno, R. (2007) 'Predicting clicks: Estimating the click-through rate for new ads', Proceedings of the 16th International World Wide Web Conference, pp. 521–530. Available at: https://doi.org/10.1145/1242572.1242643

Sutton, R.S. and Barto, A.G. (2018) Reinforcement Learning: An Introduction. 2nd edn. Cambridge, MA: MIT Press. Available at: https://www.andrew.cmu.edu/course/10-703/textbook/BartoSutton.pdf

Zhang, W., Yuan, S. and Wang, J. (2014) 'Optimal real-time bidding for display advertising', Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1077–1086. Available at: https://doi.org/10.1145/2623330.2623633

8. Appendices

Appendix A: Accessing the dashboard

The interactive dashboard was developed using the Streamlit framework and can be launched locally to explore the analytical results.

To run the dashboard, the user should first navigate to the project directory containing the dashboard script. This can be done using the terminal:

cd /path/to/project/directory

Once in the correct directory, the dashboard can be launched using the following command:

streamlit run ads_dashboard.py

Appendix B: Predictive Modelling Pipeline

The predictive modelling process was implemented using a structured pipeline that integrates preprocessing and model input preparation. Numerical variables were processed using median imputation and standardisation, while categorical variables were processed using most-frequent imputation followed by one-hot encoding. This ensures consistent transformation of features during both training and prediction.

```python
def build_preprocessor(X: pd.DataFrame) -> ColumnTransformer:
    cat_cols = [c for c in X.columns if is_object_dtype(X[c]) or is_string_dtype(X[c]) or is_bool_dtype(X[c])]
    num_cols = [c for c in X.columns if c not in cat_cols]

    numeric_pipe = Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler",  StandardScaler()),           # helps Ridge especially
    ])
    categorical_pipe = Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("ohe",     OneHotEncoder(handle_unknown="ignore")),
    ])

    return ColumnTransformer(
        transformers=[
            ("num", numeric_pipe, num_cols),
            ("cat", categorical_pipe, cat_cols),
        ],
        remainder="drop"
    )


data = df_clean.dropna(subset=[TARGET, PLACEMENT_COL]).copy()
y    = pd.to_numeric(data[TARGET], errors="coerce")
data = data.loc[y.notna()].copy()
y    = y.loc[y.notna()].copy()


leakage = [
    c for c in data.columns
    if str(c).lower() in {"ctr", "conversion rate", "conversion_rate",
                          "expected_conv_per_impression"}
    and c != TARGET
]
X = data.drop(columns=leakage, errors="ignore")

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=TEST_SIZE, random_state=RANDOM_STATE
)

pre = build_preprocessor(X_train)
```